Form 47 Mon. 8 March 2027 Reciprocal function: variations; relative position of y = x and y = x^2

1. Warming · Flash [CAL] Calculate $\frac{1}{2}$, $\frac{1}{-4}$ and $\frac{1}{0{,}1}$.

Rep. 0,5 ; −0,25 ; 10.

2. Heart · Method [CMP] Compare $\frac{1}{3}$ and $\frac{1}{5}$; what do we say about changes in the reciprocal function on $]0;+\infty[$?

Rep. $\frac{1}{3} > \frac{1}{5}$ while $3 < 5$: the opposite is decreasing on $]0;+\infty[$ (and on $]-\infty;0[$, but not on $\mathbb{R}^*$).

3. Heart + · True/false [CMP] True or false? Justify. For all real $x > 0$, $x^2 \geqslant x$.

Rep. Wrong. Counterexample: $x = \frac{1}{2}$, $x^2 = \frac{1}{4} < \frac{1}{2}$. In fact $x^2 \geqslant x$ if and only if $x \geqslant 1$ (for $x > 0$).

4. Back · Flash [Sighs] What is the front view and the top view of a cylinder placed on its base?

Rep. Face: a rectangle; on it: a disc.

5. Red thread · Prove [LIT] Show that the product of two integer pairs is a multiple of 4.

Rep. With $a = 2p$, $b = 2q$: $ab = 4pq$, multiple of 4.

Form 48 Mar. 9 March 2027 Compare f (a) *et* $f(b)$; optimization problems

1. Warming · Flash [CMP] $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.

Rep. $f(-3) = 9 > 4 = f(2)$.

2. Heart · Method [REG] $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculation, using changes.

Rep. $-2 < -1{,}5 \leqslant 0$ and f decreasing on $]-\infty;0]$: $f(-2) > f(-1{,}5)$. Control: 4 $> 2{,}25$.

3. Heart + · Optimization [MOD] $A(x) = x(10-x)$ for $x \in [0;10]$ (area of a perimeter rectangle 20). Show that $A(x) = 25 - (x-5)^2$ and deduce the maximum.

Rep. $25 - (x-5)^2 = 25 - x^2 + 10x - 25 = x(10-x)$; maximum 25 reached for $x = 5$ (side square 5).

4. Back · Flash [STA] 18 out of 24 students have more than 10 averages. Give the frequency in percentage.

Rep. $\frac{18}{24} = 75\,\%$.

5. Red wire · MCQ [EQ] The solution of $\frac{x}{3} + 1 = \frac{7}{2}$ is:

A. $\frac{15}{2}$ **B.** $\frac{9}{2}$ **C.** $\frac{21}{2}$ **D.** $\frac{5}{6}$

Rep. A. $\frac{15}{2}$

Trap: forget to subtract the 1 before multiplying by 3 ($\frac{21}{2}$).

Crossing of two qualitative variables

Form 49 11 March 2027 Crossing of two qualitative variables

1. Warming · Flash [PRO] Express $\frac{24}{60}$ in percentage.

Rep. 40 %.

2. Heart · Flash [LEC] In a class of 30 students, 18 are girls. How often are girls? Boys?

Rep. $\frac{18}{30} = 60\,\%$; 40 %.

3. Heart + · Sort [STA] Classify each variable: eye colour; baccalaureate mention (AB, B, TB) ; size in cm.

Rep. Nominal qualification; ordinal qualitative (natural order); quantitative.

4. Back · Problem [MOD] A rectangle of perimeter 20 cm has for length x. Expressing its area $A(x)$; calculate $A(3)$ and $A(5)$.

Rep. Width 10 $-x$; $A(x) = x(10-x)$; $A(3) = 21$, $A(5) = 25$ cm^2.

5. Red thread · Flash [SOL] Calculate the exact volume of a cylinder of radius 2 cm and height 5 cm.

Rep. $V = \pi \times 4 \times 5 = 20\pi$ cm^3.

Form 50 Mon. 15 March 2027 Qualitative variables; cross-table of numbers

1. Warming · Flash [CAL] Complete: 12 $+ \ldots = 30$ and $22 + \ldots = 40$.

Rep. 18 ; 18.

2. Heart · Table [LEC] Total 40 students. Girls: 22; half boarders (DP): 25; girls DP: 15. the cross-table.

	DP	External	Total
Girls	15	…	22
Boys	…	…	…
Total	25	…	40

Rep. External girls 7; DP boys 10; external boys 8; total boys 18; external total 15 (control: $7+8=15$).

3. Heart + · True/false [LEC] True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.

Rep. False. There is a natural order: it is an ordinal qualitative variable.

4. Back · Problem [CMP] Reduction A: -30% out of 80,000 Ar. Reduction B: -20 per cent then -10 per cent Is the most advantageous?

Rep. A: 56,000 Ar; B: 80,000 $\times 0{,}8 \times 0{,}9 = 57{,}600$ Ar. A is more advantageous.

5. Red thread · Method [DEM] We know that $AB^2 + AC^2 = BC^2$. What conclusion do we draw? What property?

Rep. ABC is right-angled at A : converse of the Pythagorean theorem.

Form 51 Mar. 16 March 2027 Marginal frequencies

1. Warming · Flash [PRO] Express $\frac{25}{40}$ in percentage.

Rep. 62,5 %.

2. Heart · Method [LEC] With the table in form 50: calculate the marginal frequency of girls and that of Half boarders.

Rep. Girls: $\frac{22}{\text{—}} = 55\%$; DP: 25— $= 62.5\%$ (marginal out of total staff).

Marginal frequency = total line (or column) / total number.

3. Heart + · Flash [LEC] The marginal frequency of girls is 55%. What is that of boys?

Rep. 45 %: the marginal frequencies of a variable amount to 100 %.

4. Back · Flash [SUB] Calculate $(2a-b)^2$ for $a = 3$ and $b = -2$.

Rep. $(6+2)^2 = 64$.

5. Red thread · Flash [REP] $A(-1;2)$ and $B(3;2)$. What can be said of the right (AB) ? Give AB.

Rep. Horizontal (even ordered), parallel to the axis of the apses; $AB = 3-(-1) = 4$.

Form 52 18 March 2027 Conditional frequencies

1. Warming · Flash [PRO] Give $\frac{15}{22}$ in percentage rounded to tenth.

Rep. 68,2 %.

2. Heart · Method [LEC] With the same picture: among girls, what frequency of DP? Boys?

Rep. Girls: $\frac{15}{22} \approx 68.2$ per cent; boys $\frac{10}{18} \approx 55.6\%$ (divided by total subpopulation).

3. Heart + · Distinguish [LEC] Distinguish "the frequency of girls among DPs" and "the frequency of DPs" among the girls."

Rep. $\frac{15}{25} = 60\%$ and $\frac{15}{22} \approx 68.2\%$: the reference total is not the same.

4. Back · Construction programme [chuckles] We're tracking $[AB]$ 6 cm, then center circles A and B and radius 4 cm, which intersect in C and D. Nature of $ACBD$?

Rep. A diamond: $AC = CB = BD = DA = 4$ cm (four equal sides).

5. Red thread · Flash [IVD] Is 91 first?

Rep. No: $91 = 7 \times 13$.

Trap: it is neither divisible nor divisible by 2, nor by 3, nor by 5.

Form 53 Mon. 22 March 2027 Complete a cross-table

1. Warming · Flash [PRO] Calculate 35% of 80.

Rep. 28.

2. Heart · Method [LEC] 200 clients: 60% women; 25% women and 50% men buy line. Complete the cross-table.

Rep. Women 120 of which 30 online; men 80 of which 40 online; online in total: 70.

Conditional frequencies are transferred to numbers by multiplying by the number of subpopulations.

3. Heart + · Control [LEC] A table announces "total lines: 45 $+ 55 = 100$" but "total columns: $60 + 30 = 90$". What can be concluded?

Rep. The table is incoherent: the general total must be the same by rows and columns; there is a error of entry or calculation.

4. Return · True/false [PRB] True or false? Justify. After three consecutive "stacks", the probability of obtaining "Face" to the next cast is greater than $\frac{1}{2}$.

Rep. False. The casts are independent: the probability remains $\frac{1}{2}$.

5. Red thread · Flash [GEO] ABC right-angled at A, $BC = 10$ cm, $\widehat{ABC} = 30°$. Calculate AC.

Rep. $AC = BC \times \sin 30° = 10 \times 0{,}5 = 5$ cm.

Use the sinuse: opposite side / hypotenuse.

Form 54 Mar. 23 March 2027 Actual data: file processing and filtering (AND, OR, NO)

1. Warming · Flash [LEC] In a file of 1,000 lines, 400 check A, 300 check B and 100 check A and B. How many check " A AND B »? « A OR B »?

Rep. 100 ; $400 + 300 - 100 = 600$ (it is not twice the intersection).

2. Heart · Method [LEC] With the same data: how many check "NO" A »? « A AND NO B »?

Rep. $1\,000 - 400 = 600$; $400 - 100 = 300$.

3. Heart + · Code reading [ALG] What is the final value of account ?

```
account = 0
for age in [15, 16, 17, 16, 18]:
    if age >= 16 and age < 18:
        account = account + 1
```

Rep. 3 (ages 16, 17, 16).

4. Back · Flash [NUM] Give decimal writing of $\frac{7}{8}$ then the irreducible fraction equal to 0.12.

Rep. 0,875 ; $\frac{12}{100} = \frac{3}{25}$.

5. Red thread · Formula → table [REG] $f(x) = x^2 - 2x$. Completing a table for $x \in \{0 ; 1 ; 2 ; 3\}$.

Rep. 0 ; −1 ; 0 ; 3.